Eashan M. Joshi Software Engineer

eashanjoshi4@gmail.com | (585) 202-4547 | USA | LinkedIn | Portfolio | GitHub

Summary

Innovative Software Engineer with 4+ years of expertise in building scalable, secure, and high-performance backend and frontend systems. Proficient in multiple programming languages, cloud platforms, and modern frameworks, with strong experience in microservices, API development, CI/CD, and automated testing. Adept at collaborating across teams, solving complex problems, and delivering reliable, efficient, and enterprise-grade software solutions.

Technical Skills

- **Languages & Frameworks**: Java, JavaScript, TypeScript, Kotlin, Python, SQL, HTML5, CSS3, React, React Native, Redux Toolkit, Angular, Vue.js, Spring Boot, JSP, Node.js, Express.js, Bootstrap, Tailwind CSS, FastAPI, Flask, Celery
- **Databases & Data Management**: PostgreSQL, MySQL, MongoDB, Cassandra, Redis, Apache Kafka, Apache Flink, Elasticsearch, Firebase, GraphQL, Pinecone, FAISS, Oracle SQL, PL/SQL, Azure SQL
- **Cloud Platforms & DevOps**: AWS (Lambda, API Gateway, Cognito, Fargate, CloudWatch, X-Ray, CodePipeline, CodeBuild, SNS, WAF, Shield, S3, EKS), Azure (AKS, DevOps, Function APIs), GCP, Docker, Kubernetes, Helm, Jenkins, GitLab CI/CD, GitHub Actions, Terraform, Ansible, CI/CD pipelines
- **Version Control & Collaboration Tools**: Git, GitHub, GitLab, Jira, Confluence, Slack, Trello
- **Testing & Quality Assurance**: JUnit, TestNG, Mockito, Pytest, Cypress, Selenium, Postman, Swagger, SonarQube, JaCoCo, AssertJ, JMeter
- **Architectural Patterns & Design**: Microservices, RESTful APIs, SOAP, Azure Function APIs, Event-Driven Architecture, Serverless Architecture, Domain-Driven Design (DDD), MVC, Singleton, Factory, Observer, Strategy, Repository, Dependency Injection
- **Agile & Project Management**: Agile Scrum, Kanban, Waterfall, TDD, BDD, CI/CD, Pair Programming, Code Reviews
- **Artificial Intelligence & Machine Learning**: TensorFlow, PyTorch, Scikit-learn, AWS Comprehend, NLP (spaCy, BERT, RoBERTa), OpenAI APIs (GPT-4/4o), GPT, LLaMA, Claude, LangChain / LlamaIndex, RAG, AI service integration in backend applications
- **Soft Skills**: Analytical Thinking, Problem Solving, Effective Communication, Team Collaboration, Adaptability, Time Management, Leadership, Conflict Resolution, Stakeholder Management

Professional Experience

Software Developer, *Chime* — 10/2024 – Present | Remote, USA

- Worked on Smart Finance Assistant project for Chime using AWS Lex and Polly for NLP and text-to-speech capabilities, delivering personalized banking insights. Collaborated with Product, Data Science, and Compliance teams through requirement-gathering and design workshops.
- Built secure backend microservices, TransactionService and AccountService, in Python (FastAPI, Flask) with PostgreSQL and Redis caching. Integrated UserProfile API and Payments API via RESTful endpoints for low-latency processing, optimizing performance with architecture teams.
- Created an interactive React Native dashboard using Redux Toolkit and React Query to monitor model accuracy, engagement metrics, and conversation flows. Worked with product analysts and UX teams to translate insights into improved user experiences.
- Implemented enterprise-grade authentication and authorization using AWS Cognito and JWT-based RBAC. Integrated SSO for internal tools while ensuring SOC 2, PCI-DSS, and FAPI compliance with security teams.
- Developed CI/CD pipelines with GitHub Actions and Jenkins, added static code analysis using SonarQube, and ran automated testing with Pytest, Jacoco, and Cypress. Coordinated with DevOps to maintain consistent deployment standards and code quality.
- Containerized microservices using Docker, deployed on AWS EKS with Helm charts, enabling event-driven architecture with Kafka. Implemented Prometheus and Grafana for observability, collaborating with SRE and platform teams to maintain reliability.

Software Engineer, *Morningstar India Pvt. Ltd.* — 08/2020 - 07/2022 | MH, India

- Developed enterprise financial-data systems using Python and Flask, streamlining global data workflows and reducing API response time by 35%, team up with cross-functional stakeholders to deliver, secure Azure-based backend services for Morningstar's investment research platform.
- Built Python microservices using Flask, FastAPI, Celery, and Redis for data ingestion, validation, authentication, and logging, enhancing modularity and scalability, while boosting feature delivery speed and enabling Morningstar to efficiently manage high-volume financial datasets.
- Designed and maintained diverse APIs including REST, SOAP, internal microservices, and Azure Function APIs, enabling seamless investor insights, client dashboards, and mobile integrations, improving data accuracy, availability, and reducing manual intervention by 50% across platforms.
- Developed React.js front-ends for investment-analytics dashboards and user-reporting tools, integrating with backend services. Improved UI responsiveness and performance, increasing user engagement by 30% and reducing support tickets by 25%, enhancing the overall client experience.
- Authored optimized, secure Oracle SQL and PL/SQL queries to support high-volume financial transactions. Implemented indexing, encryption, and query optimization strategies on Azure SQL, reducing latency by 45% and ensuring compliance with PII and regulatory standards.
- Led testing efforts with pytest, Postman, Selenium, and JMeter, implementing automated pipelines via Azure DevOps. Achieved 90% test coverage, reduced QA cycle time by 35%, and ensured system stability for production-ready financial analytics services.
- Collaborated with DevOps on secure deployments using Docker, Helm, and Azure DevOps, overseeing release validation, rollback planning, and infrastructure provisioning, contributing to 99.9% uptime across QA, staging, and production environments for Morningstar's platforms.

Software Engineer, *Yardi Software India Pvt. Ltd.* — 06/2019 - 07/2020 | MH, India

- Developed and maintained over 80 scalable backend components using Python and PostgreSQL, enhancing multi-tenant accounting operations for insurance, tax, and escrow workflows in high-availability production environments.
- Built RESTful APIs integrating backend systems with React-based frontends, ensuring seamless, secure, and efficient data exchange across fiduciary, compliance, and trust accounting modules for enterprise clients.
- Automated complex accounting workflows using Python, Redis, and schema-based validation logic, reducing manual operational effort by 60+ hours monthly and improving overall data accuracy and process integrity by 15%.
- Partnered with QA teams to design unit and integration test suites, increasing automated test coverage, reducing defect turnaround by 50%, and ensuring higher reliability and stability across production trust accounting systems.
- Leveraged PySpark and Kafka to optimize financial ETL processes, improving batch job reliability, data pipeline performance, and real-time reporting capabilities for large-scale periodic financial data processing workflows.

Education

Master of Science (M.S.) in Data Science 08/2022 – 05/2025
Rochester Institute of Technology, Rochester, NY, USA
Bachelor of Engineering (B.E.) in Computer Engineering 08/2016 – 05/2020
Pune Institute of Computer Technology, Pune, India

Certificates

AWS Certified Cloud Practitioner
AWS Certified Developer – Associate

Projects

CodeIntellect - Multi-Agent AI Systems Lab | Python, FastAPI, React, LLM, Docker, AWS (S3, EKS) - Demo

- Designed and deployed a LangGraph-style multi-agent system implementing the Model Context Protocol (MCP) to coordinate AI agents such as a debugger, leakage checker, and code critique, enabling automated debugging and completion of over 120 machine learning code fixes.
- Developed a FastAPI and React-based review automation platform integrated with CI/CD pipelines using GitHub Actions; containerized and deployed the system via Docker and Kubernetes (EKS), reducing manual review time by two hours per session and achieving a 4.6/5 rating.

Graph-Based Analysis of the Maven Central Ecosystem (Open-Source) | Python, Neo4j, CUDA – View Document

- Processed over 130 million dependency edges across 15 million package releases using GPU-accelerated RAPIDS cuDF and Neo4j queries to uncover ecosystem-level trends in dependency evolution, structural complexity, and version adoption patterns within the Maven Central repository.
- Analyzed dependency network growth from 2005–2024 to identify usage shifts and versioning behavior; currently extending the work to a GraphRAG framework to enable natural language querying and interactive insight generation over the entire software ecosystem graph.

Information Retrieval with Vector Embeddings for Bugzilla Reports (Open-Source) | TensorFlow, BM25 – View Document

- Built a hybrid information retrieval pipeline combining BM25, LDA with GloVe embeddings, and Siamese CNN architectures to detect semantically similar bug reports among over 100,000 Bugzilla tickets, improving top-k recall by 12% across multiple open-source projects.
- Enhanced cross-project bug retrieval and duplicate identification by optimizing embedding-based ranking models, significantly improving the efficiency of triage processes and supporting automated issue clustering for Eclipse, Firefox, and Android open-source datasets.

Publication

E. M. Joshi, M. W. Mkaouer, N. M. Vanjara, E. A. AlOmar, A. Chaaben, and M. Touati, "Tracing Dependency Dynamics in the Maven Ecosystem." IEEE International Conference on Emerging Technologies and Computing (ICETC 2025). [DOI Link] [PDF]